Print() displays output on the screen. print("Python is cool"). Commas put a space between values. They are used to put a space between values when outputting print("she/he", "Age"). print() gives a blank line. Print("="*25) gives us a multiplied string so the line of code will have 25 =. A variable is used to store data. We read = as gets. subject = "Computer". When naming variables we can use underscores but there should be no spaces. Variables are also case sensitive. We cannot use names related to Python as well like print. Every value has a type. each text is put in quotation marks and the type is string. Whole numbers are int and decimals are float. value = "Python" -> str ; number = 100 -> int ; order = 10.1 -> float We can convert between types because when a user inputs Python always enters text/string so we can convert to float or int. number = float(input("Please enter number")). We can also use + to concatenate string in Python. print("20"+"20"+"20") -> 202020 we can use f-strings to format our print. print(f"The number is: {number:>10.2f}") -> The number is +20.00. This means the number 20 will be right aligned in 10 spaces, have a sign and 2 decimal places.

NK1

Defining creates a usable block of code. Calling -> writing the functions name with () runs it. return sends a value back so it can be stored or used later. Print() displays a value on the screen. Positional values are matched to parameters by their order. keyword values are matched using parameter names. Default a parameter has a value used when no argument is provided. *args collects any number of positional arguments into a tuple. **kwargs collects any number of keyword arguments into a dictionary. local variable are created inside a function and normally can only be accessed there. use return to send a result outside a function. A docstring describes what a function does. write inside triple quotes immediately after the function definition. A module is a python file or library containing reusable code. import lets you use code from another module. if __name__ == "__main__" the block runs when a python file is executed directly, but not when imported as a module.

```
def greet():
    print("Welcome")
greet()
def multiply(a,b):
    return a*b
answer = multiply(3,4)
print(answer)
```

```
def describe(name, age=18):
    print(name, age)
describe("Lina", 20) -> positional
describe(age=19, name="Anna") keyword
describe("Mia") -> Default
```

```
def show(*args, **kwargs):
    print(args)
    print(kwargs)
show(4, 7, colour="blue", size="large")
```

WK3

We have 6 comparisons != not equal to == equal to > greater than < less than >= greater than or equal to <= less than or equal to. and both sides must be true, or at least one side should satisfy the condition. not reverses the condition. if tells python to do something only when a condition is true. else runs when the condition is false. elif means else if you use it when you have more than 2 possible outcomes. Indentation the spaces at the beginning of a line of code. It shows which statements belong together in a block of code. while loops repeat as long as the condition is True. if the condition is false it stops repeating. break = STOP the loop completely. continue = SKIP this round and go to the next round. for loops used to repeat a block of code for each item in a sequence or list. range() generates a sequence of numbers that a for loop can go through. Enumerate() allows you to loop through a sequence while getting both index and item. for loop used when you know how many times you are looping. While loop used when number of repetitions is unknown.

```
value = 40 limit = 50
print(value != limit) = True
print(value >= limit) = False
if age >= 18:
    print("You are an adult")
else: print
    print("You are not an adult")
```

```
for i in range(3):
    print(i)
```

range(5) -> 0, 1, 2, 3, 4

range(2, 10, 2) -> 2, 4, 6, 8 (start, stop, step)

```
percent = 75
if percent >= 90:
    print("A")
elif percent >= 80:
    print("B")
else:
    print("Below B")
```

~~traits = {~~

```
while True:
    password = input("Enter password")
    if password == "":
        continue
    if password == "python123":
        break
    print("Wrong password try again")
```

```
fruits = ["apple", "banana", "orange"]
for index, fruit in enumerate(fruit):
    print(index, fruit)
```

WK2

WK2